

1. Experiment 1 - Prefill/Decode Scaling

1.1 Raw Data - Run A (21 runs; 3 input settings x 3 repeats, 4 output settings x 3 repeats)

sweep	input_req	output_req	run	energy (J)	wall time (s)
input	128	64	1	155.4606	8.008
input	128	64	2	158.7099	8.0076
input	128	64	3	153.6186	8.0076
input	512	64	1	241.2782	10.0089
input	512	64	2	236.564	10.0089
input	512	64	3	240.6012	10.0092
input	2048	64	1	502.5105	28.0274
input	2048	64	2	481.142	28.0292
input	2048	64	3	480.8938	28.0285
output	512	8	1	183.1692	8.0071
output	512	8	2	180.1374	8.0076
output	512	8	3	179.6586	8.0076
output	512	64	1	221.3901	10.0091
output	512	64	2	229.3739	12.0112
output	512	64	3	220.7311	10.01
output	512	256	1	355.9769	20.0184
output	512	256	2	360.6898	20.0189
output	512	256	3	356.8552	20.0186
output	512	512	1	489.2908	28.026
output	512	512	2	505.6204	30.0293
output	512	512	3	398.5772	22.0195

1.2 Raw Data - Run B (33 runs; 5 input settings x 3 repeats, 6 output settings x 3 repeats)

sweep	input_req	output_req	run	energy (J)	wall time (s)
input	128	64	1	152.0562	7.0482
input	128	64	2	150.8309	7.0496
input	128	64	3	153.7281	7.0501
input	256	64	1	183.8467	8.0574
input	256	64	2	184.2282	8.0582
input	256	64	3	182.6025	8.0579
input	512	64	1	238.9317	10.0711
input	512	64	2	236.2654	10.0722
input	512	64	3	243.8039	10.0719
input	1024	64	1	354.3238	14.0973
input	1024	64	2	324.2488	14.1058
input	1024	64	3	302.1225	16.1435
input	2048	64	1	470.9298	27.2486
input	2048	64	2	478.0927	28.2597
input	2048	64	3	475.2503	27.2506
output	512	8	1	184.6772	8.0562
output	512	8	2	175.4261	8.0616
output	512	8	3	179.5017	8.0581
output	512	32	1	202.3694	9.0676
output	512	32	2	200.697	9.0692
output	512	32	3	200.5372	9.0675
output	512	64	1	213.4837	10.0808
output	512	64	2	219.6995	10.0793
output	512	64	3	213.536	10.0793
output	512	128	1	264.6818	13.1077
output	512	128	2	265	13.1073
output	512	128	3	268.7468	14.1166
output	512	256	1	357.2878	19.1675
output	512	256	2	363.9014	20.1765
output	512	256	3	358.6415	20.1781

output	512	512	1	448.7783	25.219
output	512	512	2	486.1695	28.2503
output	512	512	3	518.7787	30.2717

1.3 Regression Statistics and Constant Overhead

	slope (J/token)	intercept - constant overhead (J)	R-squared
Run A, output sweep (decode)	0.7716	169.83	0.9778
Run B, output sweep (decode)	0.7591	167.74	0.9978
Run B, input sweep (prefill)	0.2000	144.39	0.9850

	value
Decode marginal cost, Run A	0.7716 J/token
Decode marginal cost, Run B	0.7591 J/token
Prefill marginal cost, Run B	0.2000 J/token
Decode / Prefill ratio	3.80

1.4 Decode Share of Total Energy vs. Output Length (Run B)

actual output tokens	measured avg energy (J)	predicted decode energy (J)	decode % of total
13	179.9	9.9	5.5%
37	201.2	28.1	14.0%
68	215.6	51.9	24.1%
133	266.1	100.9	37.9%
260	359.9	197.3	54.8%
412	484.6	312.7	64.5%

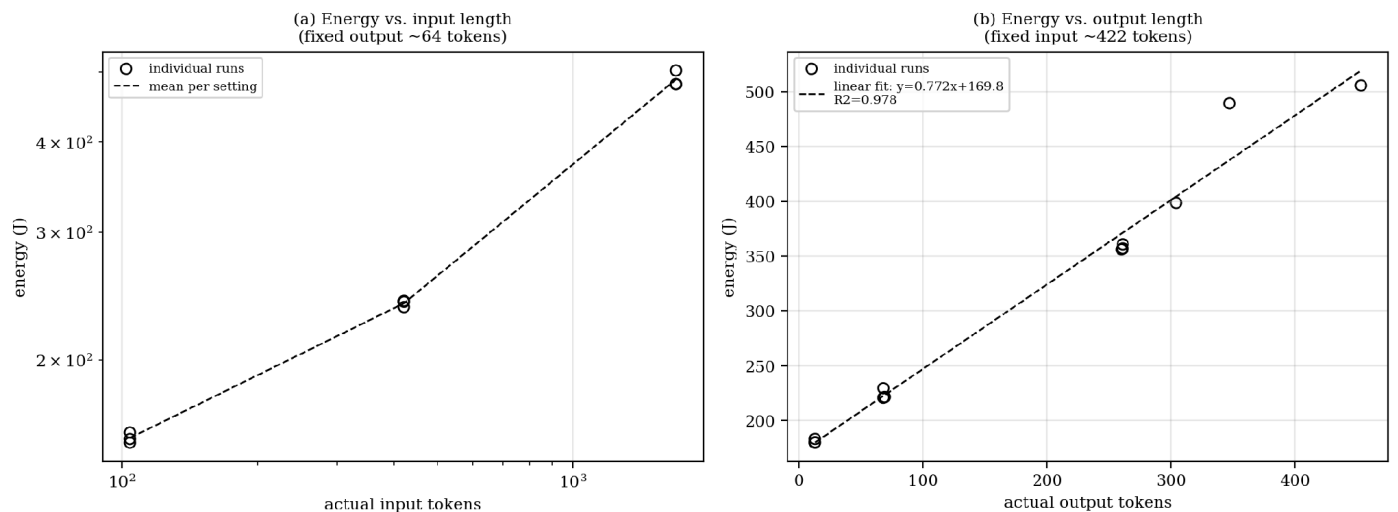


Figure B1. Energy vs. input length (a, log-log) and output length with linear fit (b), Run A (21 runs).

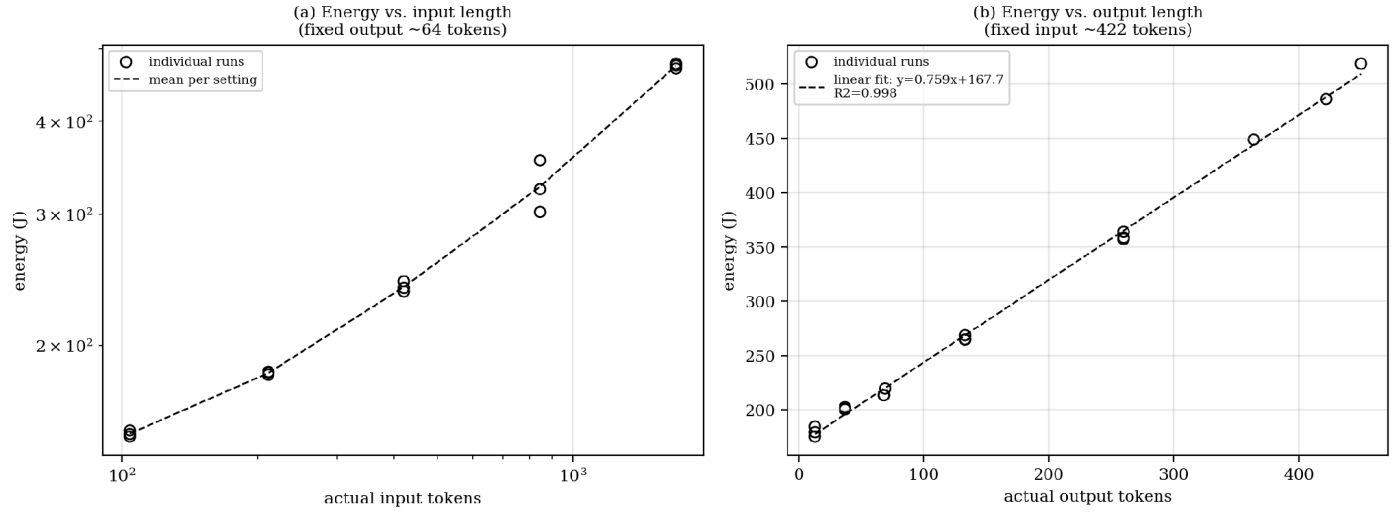


Figure B2. Energy vs. input length (a, log-log) and output length with linear fit (b), Run B (33 runs).

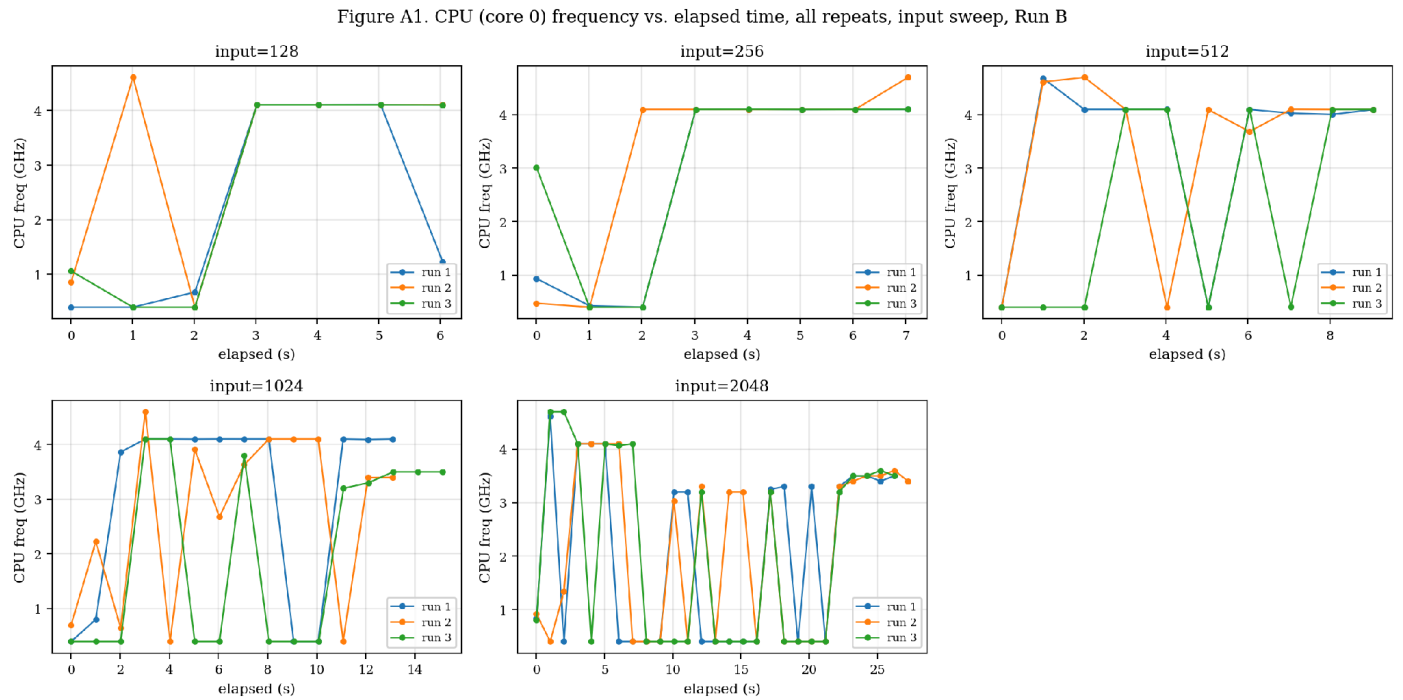


Figure A1. CPU (core 0) frequency vs. elapsed time, all repeats, input sweep, Run B.

Figure A2. CPU (core 0) frequency vs. elapsed time, all repeats, output sweep, Run B

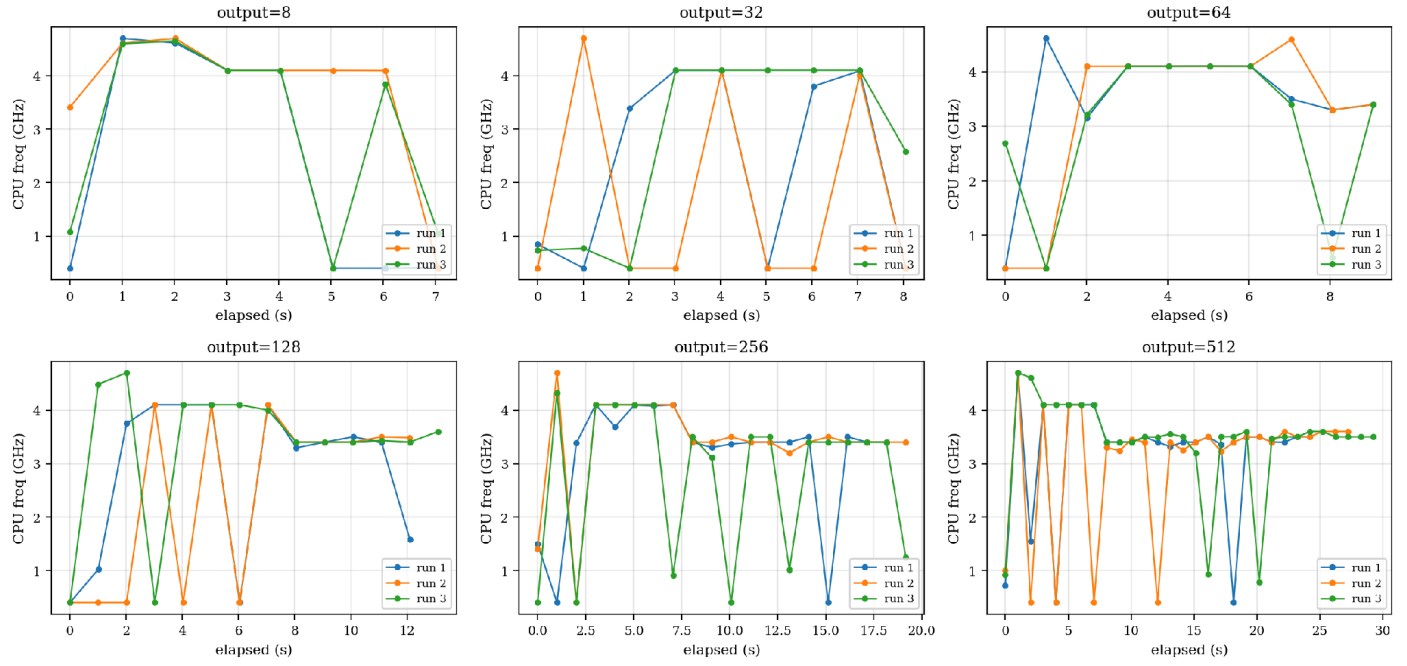


Figure A2. CPU (core 0) frequency vs. elapsed time, all repeats, output sweep, Run B.

Figure A3. Cumulative energy vs. elapsed time, all repeats, input sweep, Run B

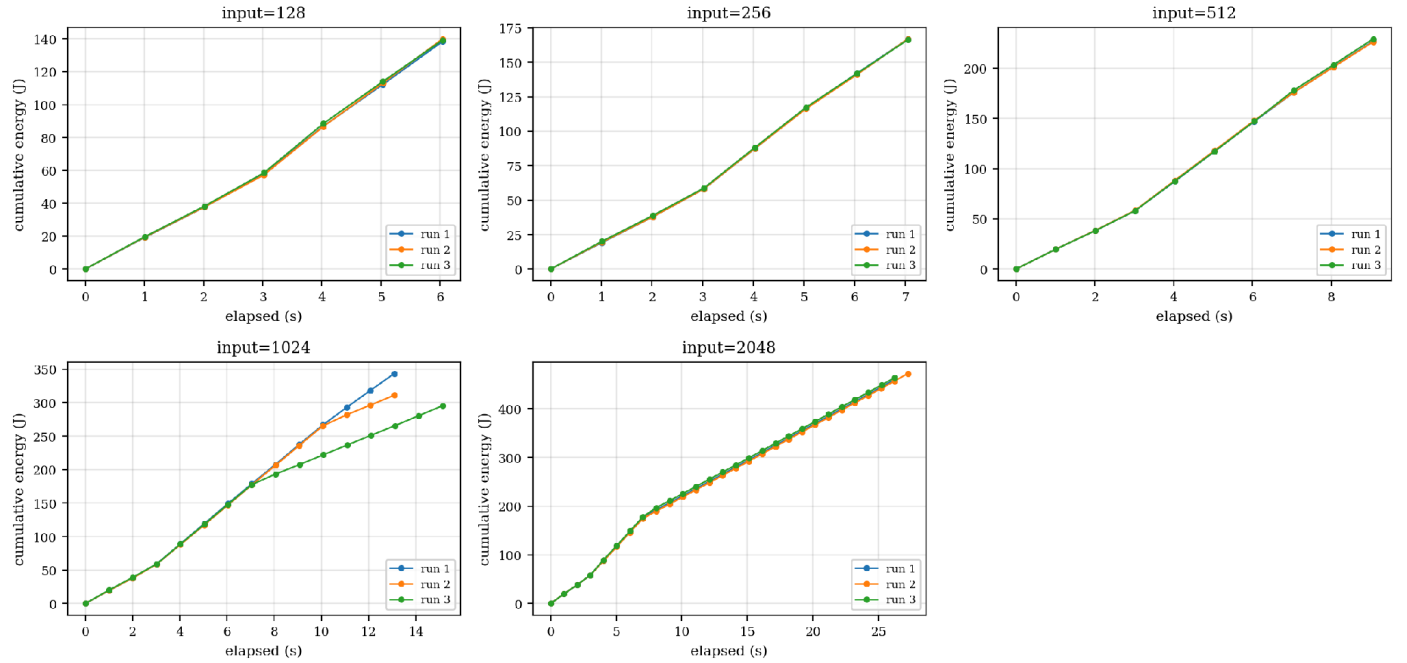


Figure A3. Cumulative energy vs. elapsed time, all repeats, input sweep, Run B.

Figure A4. Cumulative energy vs. elapsed time, all repeats, output sweep, Run B

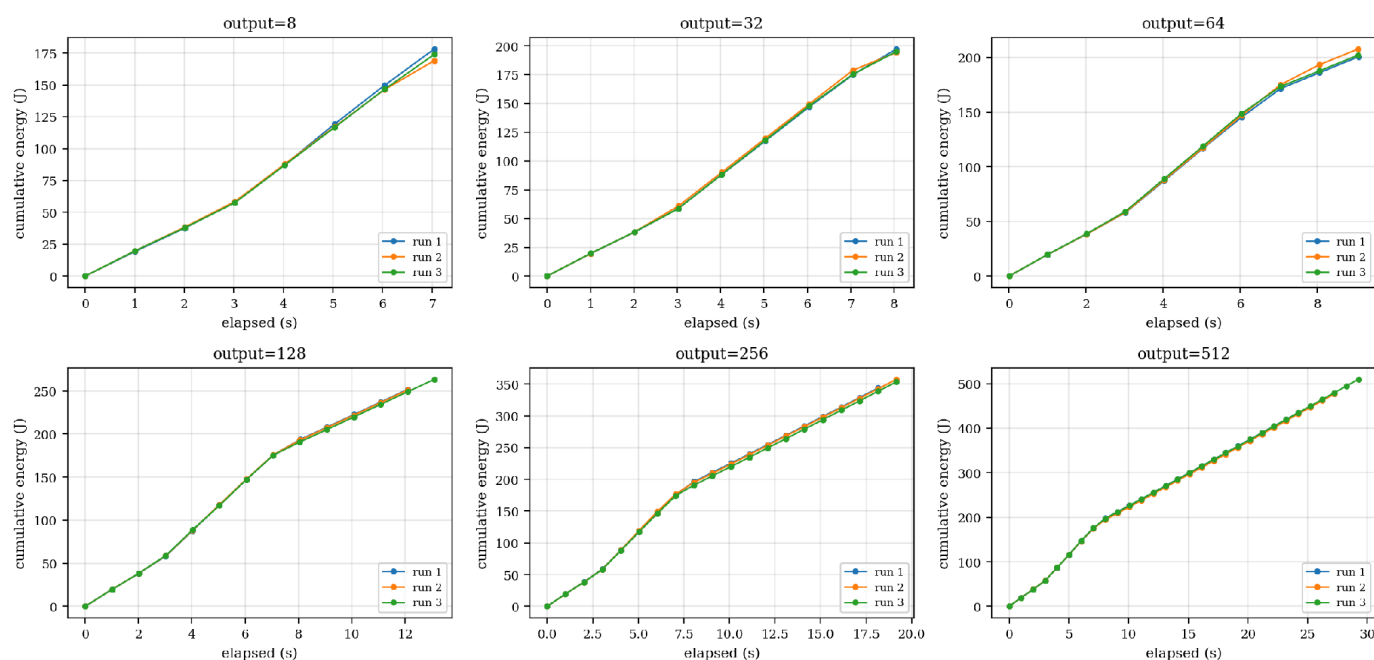


Figure A4. Cumulative energy vs. elapsed time, all repeats, output sweep, Run B.

Figure A5. Distribution of sampled CPU frequency, all runs, Experiment 1 Run B (n=464 samples)

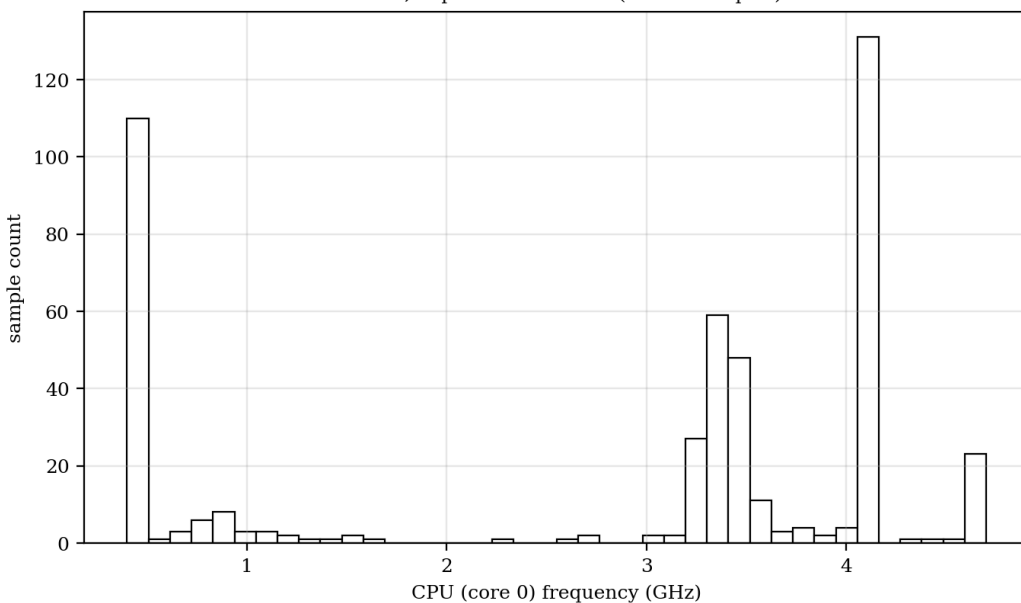


Figure A5. Distribution of sampled CPU (core 0) frequency, all runs and settings, Run B (n = 464 samples).

2. Experiment 2 - Decoding Strategy Comparison

Fixed input (~422 actual tokens). Two output-budget conditions tested: 128 tokens and 256 tokens. Strategies: greedy (temp=0), temperature (temp=0.8), top-p / nucleus (temp=0.8, top-p=0.9), top-k (temp=0.8, top-k=40). Sampler defaults (top-k=40, top-p=0.95) explicitly disabled in conditions not testing them. 3 repeats per strategy per condition.

2.1 Raw Data (24 runs)

strategy	output budget	run	energy (J)	wall time (s)	reached return?
greedy	128	1	310.88	13.0926	No
greedy	128	2	309.753	13.0951	No
greedy	128	3	310.9029	13.0938	No
temperature	128	1	315.6813	13.0985	No
temperature	128	2	273.7739	14.1138	No
temperature	128	3	276.9856	14.1167	No
top_p	128	1	277.0144	14.1161	No
top_p	128	2	274.6285	14.1153	No
top_p	128	3	272.8208	14.1184	No
top_k	128	1	273.7179	14.116	No
top_k	128	2	276.865	14.1155	No
top_k	128	3	274.8893	14.1187	No
greedy	256	1	446.6294	18.1289	Yes
greedy	256	2	446.3054	18.1337	Yes
greedy	256	3	448.6585	18.1332	Yes
temperature	256	1	385.7003	20.167	Yes
temperature	256	2	372.3329	20.174	Yes
temperature	256	3	369.2884	20.178	No
top_p	256	1	367.5491	20.1748	Yes
top_p	256	2	370.2086	20.1784	Yes
top_p	256	3	368.3637	20.1756	Yes
top_k	256	1	367.2429	20.1758	Yes
top_k	256	2	377.1157	21.1871	Yes
top_k	256	3	369.7955	20.1757	No

2.2 Summary Statistics by Strategy and Output Budget

strategy	128-tok avg energy (J)	128-tok avg power (W)	128-tok complete d	256-tok avg energy (J)	256-tok avg power (W)	256-tok complete d
greedy	310.5	23.71	0/3	447.2	24.66	3/3
temperature	288.8	20.96	0/3	375.8	18.63	2/3
top_p	274.8	19.47	0/3	368.7	18.27	3/3
top_k	275.2	19.49	0/3	371.4	18.10	2/3

Figure C1. Decoding strategy comparison, 128 vs. 256 token output budget — Llama-3.2-1B-Instruct, CPU-only

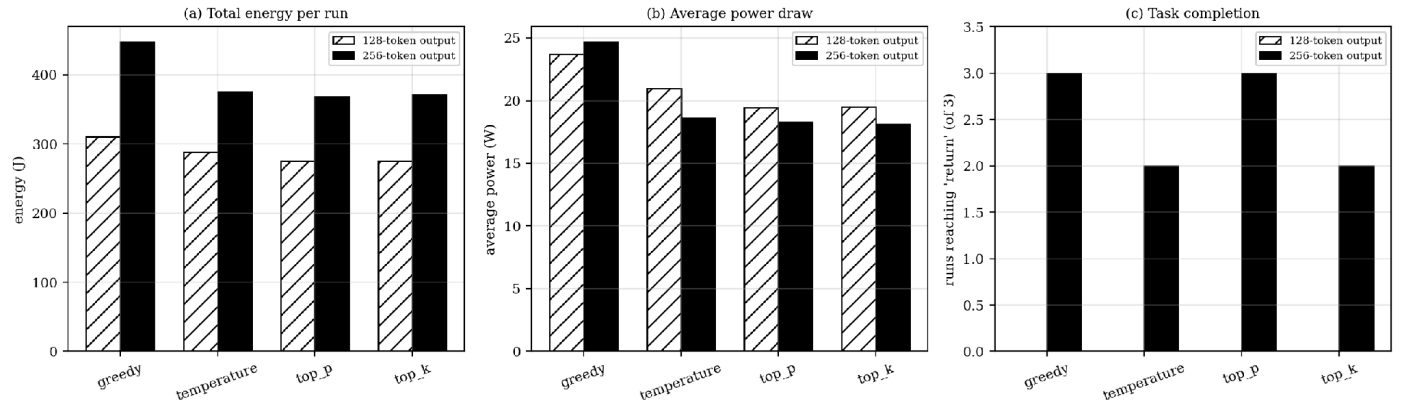


Figure C1. Total energy (a), average power draw (b), and task completion (c) by decoding strategy, at 128- and 256-token output budgets.